Jaekoo Kang

강 재 구

姜 宰 求

Phone: +1 917-497-4724
Email: jkang@gradcenter.cuny.edu
Homepage: https://jaekookang.me
Project: https://github.com/jaekookang

Education

2016 – present **Ph.D. Candidate**, *The Graduate Center, City University of New York, USA*
Speech-Language-Hearing Sciences

M.Phil., *The Graduate Center, City University of New York, USA*
Speech-Language-Hearing Sciences

2014 – 2016 **M.A. in English Linguistics** (ABD), *Korea University, Seoul, Korea*

2008 – 2014 **B.A. in English Language and Literature**, *Korea University, Seoul, Korea*

Research Interests

- Human speech production and perception
- Motor control, variability, skilled action in articulatory processes
- Native and non-native vowel production
- Speech signal processing and understanding
- Data analytics, machine learning, deep learning, computational modeling

Work Experience

2016 – present
- **Research Assistant**, *The Graduate Center, City University of New York, USA*
  Speech-Production-Acoustics-Perception Lab
  Dr. Douglas H. Whalen, advisor
- **Research Assistant**, *Haskins Laboratories, CT, USA*
  A93 Link between speech production and perception

2014 – present
- **Researcher**, EMCS/NAMZ labs
  Automatic speech recognition and synthesis

2014 – 2016
- **Research Assistant,** *Korea University, Seoul, Korea*
  Dr. Hosung Nam, advisor

Journal papers

- Lee, S., **Kang, J.**, & Nam, H. (2020). Identification of English vowels by non-native listeners: Effects of listeners' experience of the target dialect and talkers' language background. *Second Language Research*.
- Roon, K. D., **Kang, J.**, & Whalen, D. H. (2020). Effects of ultrasound familiarization on production and perception of nonnative contrasts. *Phonetica*, 1–44.
- Whalen, D. H., **Kang, J.**, Iwasaki, R., Shejaeya, G., Kim, B., Roon, K. D., Mark, K., Preston, J. L., Phillips, E., McAllister, T., & Boyce, S. E. (2019). Accuracy Assessments of

Hand and Automatic Measurements of Ultrasound Images of the Tongue. *Proceedings of International Congress of Phonetic Sciences 2019*.

Conference Proceedings

- **Kang, J.**, Nam, H., & Whalen, D. H. (2020). Estimating the "good" variability in speech production using invertible neural networks. *12th International Seminar on Speech Production*.
- **Kang, J.**, Nam, H., Chen, W., & Whalen, D. H. (2019). Benign vs. harmful variability in second language vowel production. *Proceedings of International Congress of Phonetic Sciences 2019*.
- Chen, W.-R., Tiede, M., **Kang, J.**, Kim, B., & Whalen, D. H. (2019). An electromagnetic articulography-facilitated deep neural network model of tongue contour detection from ultrasound images. *The Journal of the Acoustical Society of America*, *146*(4), 3081–3081.
- **Kang, J.**, Whalen, D. H., & Nam, H. (2018). The effect of native language on the second language vowel variability. *The Journal of the Acoustical Society of America*, *144*(3), 1868.
- Chen, W., Saltzman, E., Nam, H., & **Kang, J.** (2018). Benign vs. destructive variability in speech production: an uncontrolled manifold approach. *UConn Language Fest*.
- **Kang, J.**, Whalen, D. H., & Nam, H. (2017). Non-linear dimensionality reduction for correlated tongue measurement points. *The Journal of the Acoustical Society of America*, *141*(5), 3581–3581.
- Nam, H., **Kang, J.**, & Saltzman, E. (2017). Uncontrolled manifold method to speech production. *The Journal of the Acoustical Society of America*, *141*(5), 3584–3584.

Teaching & Talks

- **Guest lecturer** (2020). *Articulatory Phonology Fall 2020, The Graduate Center, CUNY*
  An introduction to TADA, the Haskins Laboratories Task-Dynamics Application.
- **Guest lecturer** (2018). *The Graduate Center, CUNY*
  An introduction to PsychoPy3 for designing your own online experiment
- **Guest lecturer** (2018). *Doctoral Research Spring 2018, The Graduate Center, CUNY*
  Designing behavioral experiments using Psychopy3 and analysis using Matlab

Awards & Scholarships

- **Dissertation Fellowship** (2020). The Graduate Center, CUNY
- **Doctoral Student Research Grant** (2019). The Graduate Center, CUNY
- **The Moe and Hannah Bergman Award for Conference Travel** (2017, 2019). The Graduate Center, CUNY
- **Award of the Best Student Paper** (2015). International Conference on Speech Sciences, Seoul, Korea
- **Research Assistant Scholarships** (2014, 2015). Korea University, Seoul, Korea
- **Graduation with Great Honor** (2014). Academic Affairs, Korea University, Seoul, Korea

Technical Skills

- Use of ultrasound and electromagnetic articulography for data collection and analysis, scientific data visualization (Matplotlib, Plotly), machine learning (sklearn), deep learning (TensorFlow, PyTorch), speech recognition and synthesis (HTK, Kaldi).

Programming Languages

- Python (primary), Matlab, R, Praat scripting, JavaScript, HTML, CSS and C++.